# Ghazi University

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Section: C

Lab No: 03

CamScanner

# programming fundamental Lab 3

Task 1: Assume that the following variables are defined:

```
int age;
double pay;
chas section
```

Write a single cin statement that will read input into each of These vaoiables

**Input:-**

```
#include<iostream>
using namespace std;
int main()
{
 int age=20;
 double pay= 4500.75;
 chas section='A';
 cout<<"Age:"<<age<<endl;
 cout<<"Pay:"<<pay<<endl;
 cout<<"Section:"<<section<<endl;
 cout<<"....."<<endl;

}
```

**Output**

```
Age:20
Pay:4500.75
Section:A
```

new*

```cpp

#include <iostream>
using namespace std;

int main() {
    // Task 1 - Values put kar diye hain
    int age = 20;
    double pay = 4500.75;
    char section = 'A';

    cout << "Age: " << age << endl;
    cout << "Pay: " << pay << endl;
    cout << "Section: " << section << endl;
    cout << "----------" << endl;
}
```

TAB

```
Age: 20
Pay: 4500.75
Section: A
----------

[Program finished]
```

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## Task 2

Complete the following table by writing the value of each expression in the value column according to C++ language rules

| Expression | values: |
|---|---|
| 28/4-2 | 5 |
| 6+12*2-8 | 22 |
| 4+8*2 | 20 |
| 6+17%.3-2 | 0 |
| 2+22*(9-7) | 46 |
| (8+7)*2 | 30 |
| (16+7)%.2-1 | 0 |
| 12/(10-6) | 3 |
| (19-3)*(2+2)/4 | 16 |

## Task 3

Assume a program has the following variable definition.

```
int units;
Float mass;
double weight;
```

weight = mass * units

which automatic dat type conversion will
place A-mass is demoted to an int, units re
as its and the result of mass * units is a
int

**Input**

```
#include<iostream>
using namespace std;
 int main( )
 {
  int units=5;
 float mass=2.5f;
 double weight;
 weight= mass*units;
cout<<" units(int): "<<units<<endl;
cout<<" mass(float):"<<mass<<endl;
cout<<" weight( mass*units) double me:"
<<weight<<endl;
return 0;
 }
```

**Output**

```
Unit (int): 5
mass(Float):2.5
weight (mass*unit) double
   me: 12.5
```

new*

```cpp
#include <iostream>
using namespace std;

int main(){
    int units = 5;
    float mass = 2.5f;
    double weight;

    weight = mass * units; // yahan automatic conversion hoga

    cout << "units (int): " << units << endl;
    cout << "mass (float): " << mass << endl;
    cout << "weight (mass * units) double me: " << weight << endl;

    return 0;
}
```

TAB

```
units (int): 5
mass (float): 2.5
weight (mass * units) double me: 12.5

[Program finished]
```

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sk 4:

Assume a program has the following variabl

definitions
int a+b=2, Float c=y-2;
and the following statement; a=b*c;
what value will be stored in a
A. 8.4 B. 8 C. 0 D. None of above

## Input:

Correct answer is 3.8

## C++Code:

```
#include<iostream>
using namespace std;
int main( )
  {
   int a, b=2;
  Float c= 4.2;
 a=b*c;
cout<<"b="<<b<<endl;
cout<<"c="<<c<<endl;
cout<<"b*c ="<< b*c <<endl;
 cout<<"a me Star hua:"
return 0;
```

new*

```cpp
#include <iostream>
using namespace std;

int main(){
    int a, b = 2;
    float c = 4.2;

    a = b * c; // 8.4 float se 8 int ban jayega

    cout << "b = " << b << endl;
    cout << "c = " << c << endl;
    cout << "b * c = " << b * c << endl;
    cout << "a me store hua = " << a << endl;

    return 0;
}
```

TAB

```
b = 2
c = 4.2
b * c = 8.4
a me store hua = 8

[Program finished]
```

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utput

b = 2

c = 4.2

b * c = 8.4

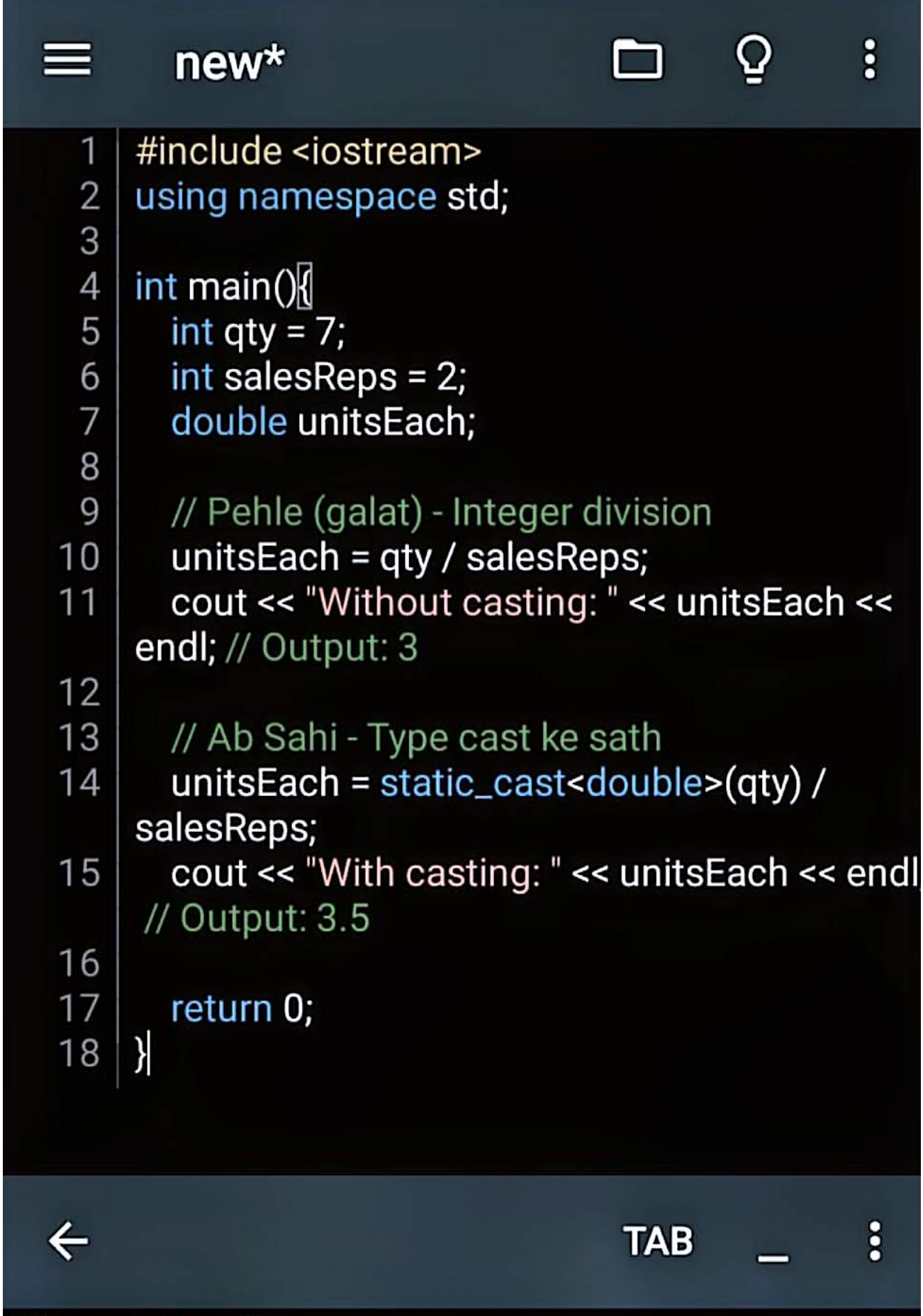
a me Hose hug = 8

## Tasks:

Assume that qty and salespeps ade both integers Use a type cast expression to rewoite the following statement so it will not longer perfom integes division

## Input:

```
#include<iostream>
using namespace std;
int main( )
{
intqty=7;
int salesReps=2;
double unitsEach;
UnitseachsqtyI/salesReps..
cout << "without costing:"<<unitsEacheendl;
UnitEachs Static-cast<double>(qty)/
SalesReps;
```




```
Without casting: 3
With casting: 3.5

[Program finished]
```

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```
cout<<" with casting": <<unitFache<<endl;
return 0;
}
```

## Output:

Without casting: 3
with casting: 3.5

## Task 6:

Write a program that can be used as a math tutor for a young student. The program should display two random numbe to be added, such as 247 + 129... .

```
#include<iostream>
using namespace std;
int main()
{
  int numb1 =rand()%.900+100;
  int num2 =rand()%.900+100;
  Int answer: num1 +num2;
cout<<" Solve this addition problem:\n\n";
 cout<<"   "<<num1<<endl;
 cout<<" + "num2<<endl;
```

```
cout<<"..."<<endl;
cin.get( );
cout<<"\nCorrect Solution:\n\n";
 cout<<"  "<<num1<<endl;
 cout<<"+ "<<num2<<endl;
 cout<<"..."<<endl;
 cout<<"  "<<answer;<<endl;
 return 0
 }
```

Output:

```
555
+978
.....
```

## Home tasks

### Task 1

Each of the following program has some errors locate as many as you can

Input

```
#Include <iostream>
using namespace std;
 int main( )
 {
```

new*

```cpp
#include <iostream>
#include <cstdlib>  // rand() ke liye
#include <ctime>    // time() ke liye
using namespace std;

int main() {
    // Random number generator ko seed karna
    srand(time(0));

    // 100 se 999 tak 2 random numbers
    int num1 = rand() % 900 + 100;  // 100-999
    int num2 = rand() % 900 + 100;  // 100-999

    int answer = num1 + num2;

    // Problem display karna
    cout << "Solve this addition problem:\n\n";
    cout << "  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "-----" << endl;

    // Pause - Student ke solve karne tak rukna
    cout << "\nPress ENTER when you are ready to check the answer...";
    cin.get();  // Enter press ka wait karega

    // Correct solution display karna
    cout << "\nCorrect Solution:\n\n";
    cout << "  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "-----" << endl;
    cout << "  " << answer << endl;

    return 0;
}
```

TAB

```
Solve this addition problem:

  555
+ 978
-------

Press ENTER when you are ready to check the answer...
```

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```
double number 1 = 15.5;
double number 2 = 10.2;
double Sum;
Sum = number 1 + number 2;
Cout << " first Number" << Number 1 << endl;
cout << 'second Number 2 << endl;
Cout << " The Sum of the two number
 is " << sum << end l;
 deturn 0;
 |
```

**Output:**

```
 7iost Number = 15.5
 Second Number. 10.2
 The sum of the two numbers is 26.7
```

**Task 2:**

**Input Code:**

```
#include <iostream>
using namespace std;
|
 int number 1 = 10;
 int number 2=4;
float quodient;
 Cout<<" Enter two numbers and I will divideh";
```

new*

```cpp
#include <iostream>
using namespace std;

int main()
{
    double number1 = 15.5;
    double number2 = 10.2;
    double sum;

    sum = number1 + number2;  // Yahan add ho raha hai

    cout << "First Number: " << number1 << endl;
    cout << "Second Number: " << number2 << endl;
    cout << "The sum of the two numbers is " << sum << endl;

    return 0;
}
```

TAB

```
First Number: 15.5
Second Number: 10.2
The sum of the two numbers is 25.7

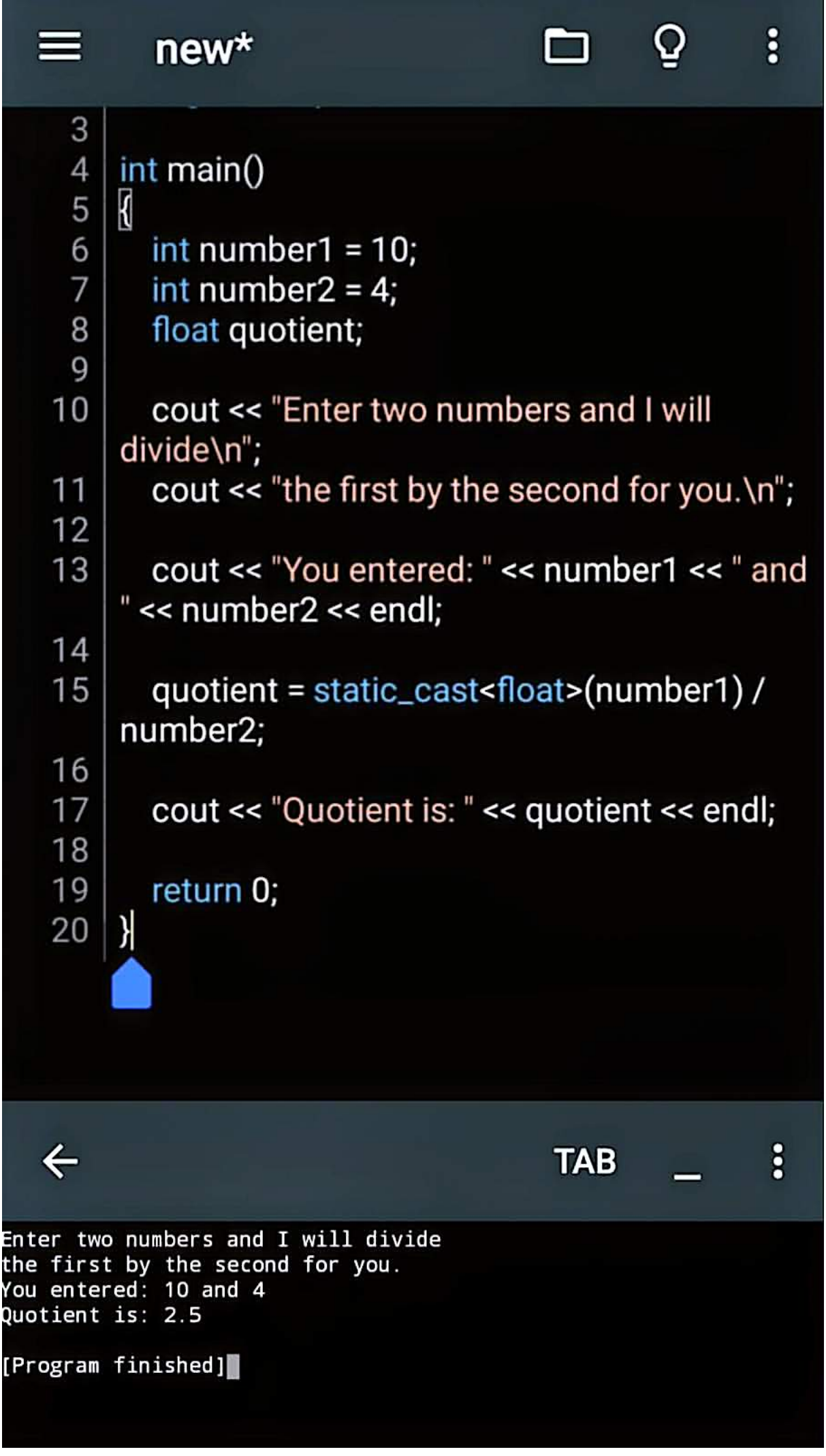
[Program finished]
```

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```
<<"the first by the second for
you.\n";
cout<<"You entaed:"<<number1<<'and "<<
number2<<endl;
quotient =static-cast<Floats>(number1)/numb
cout<<" Quotient is:"<<quotient<<endl;
return0;
}
```

**Output:**

Enter two number and I will divide the first by the second for you
You entered:10 and 4
Quotient is 2.5


new*
TAB
int main()
{
int number1 = 10;
int number2 = 4;
float quotient;
cout << "Enter two numbers and I will divide\n";
cout << "the first by the second for you.\n";
cout << "You entered: " << number1 << " and " << number2 << endl;
quotient = static_cast<float>(number1) / number2;
cout << "Quotient is: " << quotient << endl;
return 0;
}
Enter two numbers and I will divide
the first by the second for you.
You entered: 10 and 4
Quotient is: 2.5
[Program finished]

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